

Program Generation as an Evolution Operator

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Solving any problem in theory

Universal search: generate random programs and test them

Proven optimal in theory: Solomonov induction, AIXI

Issue: too slow in practice

Solving any problem in practice

Deep learning: learn representation from data

Genetic algorithms: evolve through selection and mutation

Automated theorem provers: deduce new facts using logical rules

Program networks (our approach): to be discussed

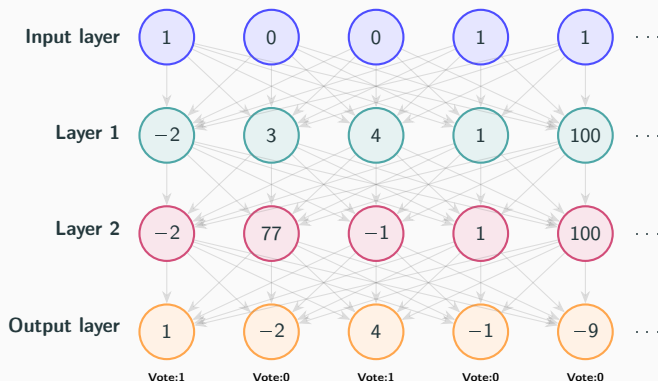
Solving any problem: a general task

General task: next bit prediction

A sequence of bits: 1,0,0,1,1,0...

Objective: predict the next bit from the previous bits

Program networks: architecture



Output Bit Prediction: 0 (3 vs 2 votes $\rightarrow$ 60% in favor of 0)

Why not use standard neural networks? delimited programs, self-reference

Program networks: integer programs

Minimal set of primitives operators:

read, x, y, z, loop, cond, 1, 2, 10, +, ×, −

Example: `(- (loop (mult 10 y) (cond 1 y
(loop (- z) x (read (- 1)))) (read 1)))`

Why choose integers as a base type?

- memory access, iteration
- easier to reason about than floating-points programs
- easier sand-boxing compared to list of integers

Program networks: training

Hill-climbing algorithm:

- *randomly* change one program node
- keep the change if it performs better (according to cross-entropy loss)

Optimizations:

- computational: parallelization, caching, dependency tracking,
- architectural: width of the network, number of layers, repeating the input, changing prior sampling probabilities, more connections, batch learning

Program networks for program networks

Can we guide changes in the network instead of using random generation?

Yes, a program network can be trained to synthesize a program for another program network.

Program networks for program networks: levels

Task 0: a sequence of bits

Level 0: train a program network for task 0

Task 1: level-0 programs as training examples

Level 1: train a program network to synthesize level-0 programs

Task 2: level-1 programs as training examples

Level 2: train a program network to synthesize level-1 programs

...

Program networks for program networks: mixing levels

Mix all the levels when training

The trained program network synthesizes programs for program networks at every level.